STUDYSPACE: A COLLABORATIVE WEB APPLICATION FOR ACADEMIC STUDENTS

Thet Oo Aung

Bachelor's thesis
Faculty of Information Technology
Czech Technical University in Prague
Department of Software Engineering
Study program: Informatics
Specialisation: Software Engineering
Supervisor: Ing. Jan Blizničenko, Ph.D.
June 24, 2026

FACULTY
OF INFORMATION
TECHNOLOGY
CTU IN PRAGUE

Assignment of bachelor's thesis

Title:	StudySpace: A Collaborative Web Application for Academic Students
Student:	Thet Oo Aung
Supervisor:	Ing. Jan Blizničenko, Ph.D.
Study program:	Informatics
Branch / specialization:	Software Engineering 2021
Department:	Department of Software Engineering
Validity:	until the end of summer semester 2026/2027

Instructions

The goal of this thesis is to design and implement a functional prototype of a web application that combines course materials management and student communication into a single environment.
The application aims to reduce the need for switching between different tools by linking study materials directly to peer discussions.

Key Functional Requirements:
1. Course Administration Module: An interface for instructors to create course structures, upload lecture materials, and handle student enrollments.
2. Content Extension System: This feature allows students to clone course materials into private workspaces for personal note-taking and group collaboration, or to propose their modifications to the instructor. The instructor can manually review them and choose to showcase the student's extended version alongside the official content for the entire class.
3. Contextual Messaging System: A communication tool that allows students to "anchor" questions to specific course materials. By typing a shortcut symbol (such as @), students can select a PDF or files from the course materials to include in their message. This creates a clickable link, allowing others to jump directly to the selected document for quick reference and support.
4. AI-Based Content Querying: Integration of the APIs from Large Language Models (e.g. OpenAI, Gemini) to allow users to query summaries or explanations based on the

---

Electronically approved by Ing. Michal Valenta, Ph.D. on 11 February 2026 in Prague.

context of the loaded study materials.
5. Access Control: A secure authentication/ authorization system to differentiate permissions depending on the roles: eg. Students (view/copy/discuss) and Instructors (manage content/approve merges).

Follow these steps:
1. Analyze Requirements: Define the functional requirements for StudySpace and analyze the specific needs of the target academic audience through user research.
2. Current Solutions Analysis: Research existing solutions (Learning Management Systems like Moodle, communication tools like Discord/Teams) to identify integration gaps and opportunities for improvement.
3. UI Design: Create wireframes to visualize the prototype's flow for a better user experience.
4. System Architecture: Design the database schema and API structure. Research and select the appropriate technology stack (programming languages, frameworks, libraries, tools), specifically addressing the data model required for personal content copies, merge proposals and contextual message anchoring.
5. Implementation: Prioritize the development of a functional prototype of the application with the frontend and backend components, integration of the LLM API, and implementation of all core features.
6. Testing: Perform functional and usability testing to ensure system stability and completeness of the core features.
7. Documentation: Write technical documentation for deployment and user guides for both students and instructors.

Citation of this thesis: Aung Thet Oo. *StudySpace: A Collaborative Web Application for Academic Students*. Bachelor's thesis. Czech Technical University in Prague, Faculty of Information Technology, 2026.

I would like to thank my supervisor, Ing. Jan Blizničenko, Ph.D., for his guidance throughout the preparation of this thesis. I would also like to thank my family for their continuous support and encouragement.

Declaration

I hereby declare that this bachelor's thesis entitled "StudySpace" was written by me independently and that I have used only the sources listed in the bibliography. I declare that I have not submitted this thesis or any part thereof for the purpose of obtaining any other degree or qualification.

I acknowledge that my thesis is subject to the Act No. 121/2000 Coll., the Copyright Act of the Czech Republic, and that Czech Technical University in Prague has the right to conclude a license agreement on the use of this thesis as specified by this Act.

In Prague on June 24, 2026

Abstract

This bachelor's thesis describes the design and implementation of a prototype of StudySpace, a web application that allows students and instructors to manage course materials and collaborate in one place. The application addresses the problem of students having to switch between multiple separate tools for materials, communication, and assignments.

The thesis presents an analysis of existing learning management systems and system requirements, followed by the design and implementation of key features: course management, a system where students can clone a workspace, add their own notes, and contribute the updated document back to the original course page for the instructor to review, messages linked to specific parts of the content, and an AI assistant for querying study materials. The backend is built with Java and Spring Boot, the frontend with Next.js and React, using PostgreSQL for data storage. Deployment is automated via CI/CD pipelines to modern cloud platforms (Vercel, Render). The system is verified with unit, repository, and integration tests.

Keywords web application, collaboration, React, Spring Boot, Next.js, PostgreSQL, REST API, CI/CD, WebSocket

Abstrakt

Tato bakalářská práce popisuje návrh a implementaci prototypu webové aplikace StudySpace, která umožňuje studentům a vyučujícím spravovat studijní materiály a spolupracovat v jednom prostředí. Aplikace řeší problém, kdy studenti dnes používají několik různých nástrojů najednou pro materiály, komunikaci i odevzdávání úkolů.

Práce zahrnuje analýzu existujících systémů pro správu výuky a následnou implementaci klíčových funkcí: správu kurzů, systém, ve kterém si student může naklonovat studijní pracovní prostor, přidat vlastní poznámky a upravený dokument přispět zpět na původní stránku kurzu ke kontrole vyučujícím, zasílání zpráv navázaných na konkrétní části obsahu a dotazování materiálů pomocí umělé inteligence. Backend je postaven na Java a Spring Boot, frontend na Next.js a React, databáze je PostgreSQL. Nasazení využívá moderní CI/CD pipelines do cloudového prostředí (Vercel, Render). Systém je otestován jednotkovými, repozitářovými a integračními testy.

Klíčová slova webová aplikace, spolupráce, React, Spring Boot, Next.js, PostgreSQL, REST API, CI/CD, WebSocket

Contents

List of Figures

List of Tables

List of Code listings

List of abbreviations

API	Application Programming Interface
CS	Computer Science
DB	Database
DTO	Data Transfer Object
ER	Entity-Relationship
HTTP	Hypertext Transfer Protocol
JPA	Java Persistence API
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
LMS	Learning Management System
MVC	Model-View-Controller
ORM	Object-Relational Mapping
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
REST	Representational State Transfer
SPA	Single Page Application
SQL	Structured Query Language
SSR	Server-Side Rendering
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience
UML	Unified Modeling Language

Chapter 1
Introduction

Imagine a student preparing for an exam at the Faculty of Information Technology at CTU in Prague. She has a question about a specific diagram on slide 14 of the lecture notes uploaded to the faculty course system. To ask her peers, she must switch to Discord, copy the name of the file from the course page, paste it into a message, and hope that her classmates can locate the same slide quickly enough to give a useful answer. The question floats away in the message history by the next morning, disconnected from the material it was about and invisible to students who enrol in the course the following semester. This scenario, repeated across every course and every cohort, illustrates the fundamental problem that StudySpace is designed to address: the fragmentation of academic tools that forces students and instructors to operate across multiple disconnected platforms.

1.1 Motivation

Modern higher education relies on a combination of platforms that rarely communicate with one another. At a typical faculty, course materials reside in a learning management system such as Moodle or the faculty course portal, while student collaboration happens informally on Discord, WhatsApp, or Microsoft Teams. Code assignments and grading are handled by yet another system, and AI tools such as ChatGPT are used in isolation without any link to the actual curriculum content. The consequence of this fragmentation is that information ends up scattered: a useful correction that a student discovers during self-study has no path back into the course, a question about a specific document carries no reference to the document itself, and knowledge generated by one cohort is invisible to the next.

StudySpace addresses these gaps by combining four capabilities in a single environment. Instructors need a structured way to organise and publish course materials and to manage which students can access them (F1). Students, rather than passively consuming that content, need a mechanism to clone a workspace, enrich it with their own notes, and contribute improvements back to the original course page for the instructor to review and optionally publish (F2). When students want to ask a question about a specific document, they need a messaging channel that can attach a direct reference to that document, so other readers can follow the link and land on the relevant material immediately (F3). Finally, students need an AI assistant that queries the actual course content rather than operating on general knowledge alone, grounding answers in the material the instructor has provided (F4). All of these capabilities require a role-based access model that distinguishes instructors from students at the API level, preventing students from modifying course materials they do not own (F5).

1.2 Structure

Chapter 2 defines the target audience and their pain points, establishes the functional and non-functional requirements derived from the motivation above, examines existing platforms and their limitations, and describes the planned system and logical architecture. Chapter 3 translates the analysis into concrete technical decisions, covering the domain model, the technology choices and the rationale behind each, the database schema, and the user interface design. Chapter 4 documents the implementation of the five core features, presents the code listings that illustrate the most important design choices, and describes how the system was tested. Chapter 5 summarises the outcomes relative to the goals stated in this chapter, reflects on the known limitations of the prototype, and proposes directions for future work.

Chapter 2

Analysis

This chapter establishes the foundation for all subsequent design and implementation decisions. It begins by defining the problem and the target audience whose needs motivated this project, then describes the intended architecture at a conceptual level before any technology is named. The formal functional and non-functional requirements are stated next, followed by a use-case view grouped by actor. The chapter closes by examining existing platforms and identifying the integration gap that none of them currently fills.

2.1 Problem Statement and Target Audience

The primary target audience of StudySpace consists of two groups whose workflows are closely coupled: *students* and *instructors* at a higher-education faculty. Students are the consumers and potential contributors of course content. Their typical lifecycle within a single course involves browsing uploaded materials, asking questions about specific documents, annotating or extending study notes independently, and occasionally contributing improvements that could benefit their peers. Instructors are the publishers and curators of that content. Their lifecycle involves creating a course structure, uploading materials into named sections, managing which students are enrolled, and reviewing any student contributions before deciding whether to make them visible to the whole class.

The Faculty of Information Technology at CTU in Prague serves as the representative context. A student's daily workflow involves switching between multiple systems: the faculty course portal or Moodle for lecture slides and reading assignments, Microsoft Teams or SharePoint for live sessions and recordings, Progtest or GitLab for programming assignments and grading, and Discord, WhatsApp, or email for informal peer discussion. This constant context-switching leads to scattered information, loss of peer-generated knowledge, and extra effort spent locating the right tool for a given task. Discussions about a specific lecture slide happen in a general chat channel with no link to the actual material. Student notes and annotations remain in personal files and never contribute back to the course. There is no structured way for a student to propose improvements to lecture materials, and AI tools are used in isolation without any connection to the course content.

StudySpace aims to unify course management, student collaboration, contextual discussion, and AI assistance into a single platform, so that the interactions between these activities are preserved and visible to all participants.

2.2 Architecture Analysis

This section justifies each major technology choice by comparing the available alternatives against the requirements established in Chapter 2. The goal was to select a stack that is well suited for the required features, has strong community support and ecosystem maturity, and is practical to use within the scope of a bachelor thesis prototype.

2.2.1 Frontend Framework

The core capabilities of the system (such as course administration, content cloning, contextual messaging, and AI assistance) require a responsive and component-rich user interface that manages complex shared state across a course-material viewer, a chat panel, and an AI assistant panel rendered simultaneously on the same screen.

React 19 [1] is a declarative UI library maintained by Meta. Its component model encourages decomposing the interface into small, reusable pieces, each responsible for a single part of the view. The unidirectional data flow makes state changes predictable and easier to trace during debugging. React's large ecosystem provides ready-made solutions for drag-and-drop panels, rich-text editing, and real-time subscriptions, all of which are needed for StudySpace.

Vue.js is a progressive framework with a smaller bundle size and a gentler learning curve. For StudySpace, however, the Vue ecosystem is shallower for complex, enterprise-grade components, and the TypeScript integration, while improving, is less seamless than in React.

Angular is a full-featured framework backed by Google that includes dependency injection, a router, and a form library out of the box. The overhead of Angular's concepts, decorators, and module system exceeds what is justified for a single-team prototype of this scope.

React was selected because its component model aligns directly with the StudySpace screen layout (course panel, chat sidebar, AI panel as discrete components), its TypeScript support is first-class, and its ecosystem provides the resizable panel, rich-text input, and WebSocket hook primitives needed for the content extension and contextual messaging features without significant additional integration effort.

React is used through the **Next.js 15** [2] framework, which provides file-based routing via the App Router, server-side rendering where applicable, and built-in image optimisation. Compared to a plain Vite or Create React App setup, Next.js removes boilerplate routing configuration and makes the project structure self-documenting through the directory layout. Styling is handled by **Tailwind CSS** [3] combined with **Shadcn/ui** [4] and **Radix UI**, which provide accessible, unstyled component primitives that Tailwind classes are applied to. State management across the application uses React's native **Context API** and custom hooks, providing a lightweight, predictable data flow for all API interactions without the overhead of external state libraries.

2.2.2 Backend Framework

The system requires robust access control with role-based authorisation. Furthermore, the content extension workflows and AI assistance feature involve complex transactional operations over relational data. These requirements favour a statically typed language with mature transaction management and security libraries.

Java 21 with Spring Boot 3 [5, 6] provides a statically typed, object-oriented foundation with a mature ecosystem. Spring Boot's convention-over-configuration approach applies sensible defaults automatically, embeds a Tomcat web server, and offers first-class integration with Spring Security, Spring Data JPA, Spring WebSocket, and Springdoc OpenAPI — all of which are essential for StudySpace. Virtual Threads, introduced in Java 21, improve the throughput of concurrent request handling without the complexity of reactive programming.

Node.js with NestJS or Express excels in I/O-bound tasks due to its event-driven architecture. However, its single-threaded event loop can be a bottleneck for the multi-step transactional operations required by the Content Extension System, and the ORM ecosystem does not reach the integration depth and stability of Spring Data JPA.

Python with Django or FastAPI is excellent for rapid prototyping and AI integration. Python's dynamic typing, however, makes maintaining large-scale backend codebases more challenging without strict tooling, and its Global Interpreter Lock can become a bottleneck for highly concurrent CPU-bound tasks.

Java 21 with Spring Boot 3 was selected because the controller-service-repository layered model maps directly onto the StudySpace domain structure, Spring Security provides JWT-based authentication and method-level authorisation out of the box (directly addressing the access control requirements), and the mature transaction management ensures referential integrity across the complex clone and proposal approval operations required by the content extension workflow.

2.2.3 Database and Vector Store

The application's data model is inherently relational: users own workspaces, workspaces belong to courses, clone records track inheritance, and proposals reference both student materials and instructor courses. The content extension system requires ACID transaction guarantees across multi-entity writes.

PostgreSQL 17 [7] is a mature, open-source relational database management system (RDBMS) that follows the SQL standard and supports complex queries, transactions, and foreign key constraints. It integrates seamlessly with the Java backend through the JDBC driver, and higher-level access is provided by Spring Data JPA with Hibernate as the ORM provider.

MongoDB (NoSQL document store) and **Cassandra** (wide-column store) are optimised for unstructured or semi-structured data and horizontal scalability. However, they lack strict schema enforcement and standard join operations. Using a document store would require significant denormalisation, duplicating data to avoid application-level joins, which introduces the risk of inconsistency when a Course Material is approved and must be reflected in both the workspace and the course tables atomically.

PostgreSQL was selected because the StudySpace data model is fundamentally relational, ACID transactions are required for the clone and approval workflows, and foreign-key constraints enforce referential integrity between proposals and their source materials. Additionally, the AI assistant leverages the `pgvector` extension to provide similarity search over document embeddings directly within the same database, eliminating the need for a separate dedicated vector store.

2.2.4 Authentication and Authorisation

The access control system requires stateless authentication suitable for a REST API consumed by a single-page application, with role-based restrictions enforced on the server side.

JSON Web Tokens (JWT) with Spring Security [8] enables stateless authentication: after login, the server issues a signed token that the client stores and sends with every subsequent request. The server validates the token signature without consulting a session store, which simplifies horizontal scaling.

Server-side sessions with Spring Session require a shared session store (such as Redis) across instances and add network latency to every request. For a stateless REST API served to a browser-based SPA, this overhead is unnecessary.

OAuth 2.0 only (without local credentials) would prevent users without a Google or GitHub account from registering, which is inappropriate for a faculty application where students may not use third-party providers for academic accounts.

JWT with Spring Security was selected because it achieves stateless session management (directly satisfying NFR-02), requires no external session store, and Spring Security's filter chain integrates the JWT validation and role check into a single, reusable `OncePerRequestFilter` implementation. Google OAuth 2.0 login is supported as an optional alternative authentication path alongside the primary local credential flow.

2.2.5 Real-Time Communication

The Contextual Messaging System requires peer-to-peer discussion to occur in real time. This necessitates bidirectional event delivery to all connected participants, ruling out traditional request-response and simple polling-based approaches.

WebSocket with STOMP over SockJS [9] layers a publish-subscribe model on top of the raw WebSocket connection. Spring Boot registers a broker endpoint and routes messages to per-session topics, allowing the server to push participant-list updates and activity events to all subscribers without the clients polling.

Server-Sent Events (SSE) provide a unidirectional push channel from server to client. They lack the bidirectional subscription model needed for clients to send activity events (such as a hand-raise) back to the server over the same connection.

Long Polling achieves real-time delivery but requires each client to maintain a pending request and re-establish it after every response, producing significant per-client overhead on the server.

WebSocket with STOMP over SockJS was selected because the bidirectional subscription model covers both server-to-client pushes (participant list updates, activity events) and client-to-server messages (hand raises) over a single connection, and Spring Boot's built-in STOMP message broker requires no additional infrastructure.

2.2.6 Quality Assurance

Ensuring the reliability of the StudySpace web application requires a robust quality assurance strategy, encompassing both isolated component testing and full-system end-to-end (E2E) verification. The selection of appropriate testing frameworks must account for the project's reliance on the Next.js framework, the need for rapid developer feedback, and the complexity of real-time browser interactions.

2.2.6.1 Unit and Integration Testing

Unit and integration tests verify individual application components, such as the contextual chat panel or the course material upload logic, in isolation from the full backend server. The two primary candidates for the React ecosystem are Jest and Vitest.

Jest [10] is a mature, widely adopted testing framework maintained by the open-source community. It provides a comprehensive set of assertion libraries and mocking utilities out of the box, and Next.js offers built-in configuration wrappers for it. However, Jest relies on a traditional Node.js execution model and requires complex transformations to handle modern ECMAScript modules (ESM) and TypeScript natively. As the StudySpace codebase grows, this transformation overhead can significantly slow down test execution.

Vitest [11] is a newer framework built on top of Vite. It is designed to be API-compatible with Jest, allowing developers to use identical syntax for assertions and mocks, but it uses Vite's highly optimised build pipeline for module resolution and transformation. This results in significantly faster test execution and near-instantaneous Hot Module Replacement (HMR) during test-driven development. In the context of StudySpace, where frequent iterations on the user interface and complex state management hooks are required, minimising the feedback loop is critical. Although Next.js uses Webpack and Turbopack rather than Vite for its production builds, Vitest can be configured to understand the Next.js environment seamlessly.

Given the performance advantages of its modern execution pipeline and its seamless compatibility with the React Testing Library, Vitest was selected as the unit and integration testing framework. The speed improvements directly support the rapid development cycles required for the project, while the familiar Jest-like API minimises the learning curve.

2.2.6.2 End-to-End Testing

End-to-end testing simulates real user behaviour by driving an actual web browser, ensuring that the frontend, backend, and database interact correctly. This is particularly important for StudySpace, as features like the Merge Proposal workflow and the real-time Contextual Messaging System involve complex cross-tier interactions. Cypress and Playwright are the leading solutions for modern web applications.

Cypress [12] offers a highly visual, interactive test runner that executes directly within the browser alongside the application under test. This architecture provides excellent debugging capabilities and a shallow learning curve. However, Cypress traditionally struggles with testing multi-tab workflows and cross-origin navigations, which can be problematic when testing OAuth flows or interactions that open course materials in new tabs. Furthermore, its execution speed is relatively slow when running large test suites in headless Continuous Integration (CI) environments.

Playwright [13], developed by Microsoft, takes a different architectural approach by communicating with browsers over the Chrome DevTools Protocol (CDP). It natively supports parallel test execution across multiple browser engines (Chromium, WebKit, and Firefox) and handles multi-tab, multi-origin scenarios effortlessly. Playwright also includes a powerful trace viewer that captures DOM snapshots, network requests, and console logs for every step of a failed test, significantly easing debugging without requiring the interactive test runner to be open.

For StudySpace, the ability to test complex user journeys efficiently is paramount. The parallel execution model of Playwright drastically reduces the time required to run the full E2E suite compared to Cypress. Additionally, its robust handling of modern web APIs and multi-context scenarios ensures that the real-time communication channels and file viewing features can be tested reliably. Consequently, Playwright was chosen as the end-to-end testing framework for the project.

2.3 Functional Requirements

The following requirements are derived directly from the problem statement and the target audience description in section 2.1. Each requirement is identified by an ID that is referenced throughout the remainder of the thesis.

FR-01 (Course Administration): An Instructor must be able to create, update, and delete courses and organise their content into named sections, while a Student must be able to browse and read any course they are enrolled in.

The Instructor creates a course with a title and description and controls whether it is published and visible to students. Within a published course the Instructor defines *sections*, each with an ordered index that determines the display sequence. Course Materials (PDF documents, slide sets, video files, or images) are uploaded into sections; the backend infers the file type from the MIME type and extension and stores it alongside the file URL so the frontend can render the appropriate viewer without an additional request. Enrollment is managed per student: the Instructor can activate or drop individual students, and a student sees only the courses in which their enrollment status is `ACTIVE`.

FR-02 (Content Extension System): A Student must be able to clone a course into a Private Workspace, add or modify materials independently, and submit a Merge Proposal for the Instructor to review and optionally publish.

When a student clones a course, the backend performs a deep copy of the course structure into a new `StudentWorkspace`: every section and every Course Material is mirrored as a `WorkspaceSection` and a `WorkspaceMaterial`. Cloned materials carry the flag `isReference = true`, meaning they point to the original file URL rather than a new copy, which saves storage. The student may then upload additional materials into the Private Workspace. When satisfied, the student submits a Merge Proposal, which the Instructor reviews in a dedicated queue. On approval, the backend copies the student's file into the course storage and publishes it as a new Course Material credited to the contributor.

FR-03 (Contextual Messaging System): Users must be able to post messages in the workspace chat and attach a direct reference to a specific Course Material, so that other users can open the material from within the message.

The chat panel is embedded in the workspace view alongside the material list. Users compose messages in a rich-text input and can type `@` followed by a material name to trigger an autocomplete dropdown. Selecting a material embeds a non-editable tag chip in the message. When the message is submitted, the chip is serialised into a portable token of the form `@[id:title]`, which is stored as the message text. On render, the token is parsed back into a clickable badge, which the user clicks to download or open the referenced Course Material. This Contextual Anchor persists in the chat history and remains accessible to every enrolled student who opens the workspace later.

FR-04 (AI-Based Content Querying): Users must be able to ask natural-language questions about the content of the currently open workspace and receive answers grounded in that content.

The AI panel is a collapsible sidebar within the workspace view. When the user submits a question, the frontend sends the question together with the URL of the selected Course Material to the backend. The backend extracts the plain text from the document, constructs a structured prompt that includes a system instruction, the document context truncated to 10 000 characters, and the user's question, then forwards the prompt to the Large Language Model (LLM) API. The response is streamed back to the frontend and displayed with a typewriter animation. Conversation history is preserved in the database through a hybrid memory model: a rolling buffer of recent messages and a compressed long-term summary, ensuring that the AI can reference earlier turns in a session even after the short-term buffer overflows.

FR-05 (Access Control): The system must recognise two roles, Student and Instructor, and enforce role-based restrictions at the API level so that students cannot perform instructor-only operations.

Role assignment happens at registration and is stored as the `role` field of the `User` entity. The authentication flow issues a signed JSON Web Token (JWT) upon successful login; the token encodes the user's email and is valid for ten hours. Every subsequent request must carry the token in the `Authorization: Bearer` header. The `JwtAuthenticationFilter` intercepts each request, extracts and validates the token, and populates the Spring Security context. Instructor-only endpoints are protected with `@PreAuthorize` annotations or explicit ownership checks in the service layer so that a student account receives an HTTP 403 response if it attempts a restricted operation.

2.4 Non-Functional Requirements

The following non-functional requirements define the quality constraints the system must satisfy regardless of the specific feature being used.

- **NFR-01 (Usability)**: The interface should be straightforward enough for students with no prior training to use without a manual.
- **NFR-02 (Security)**: All endpoints except login and registration must require a valid JWT token. Role-based access must be enforced on the server side.
- **NFR-03 (Performance)**: Course and workspace pages should load within three seconds under normal conditions.
- **NFR-04 (Maintainability)**: The backend and frontend must be clearly separated and independently deployable. The project structure should follow established conventions for its respective framework.
- **NFR-05 (Deployability)**: The frontend and backend must be clearly separated and independently deployable via automated CI/CD pipelines to modern cloud platforms. Local development should still be achievable with a single Docker Compose command.

2.5 Use Cases

The system has two types of actors: *Student* and *Instructor*. The use cases below describe the main interactions each actor has with the system, grouped by role. Pre-conditions are stated for the non-trivial cases that have dependencies on prior system state.

Instructor Use Cases

- **`Create course`**: The Instructor creates a course with a title and description and sets its published status.
- **`Upload Course Material`**: Pre-condition: the course and at least one section exist. The Instructor selects a file, assigns a title, and uploads it into a named section. The backend infers the file type and persists the file URL.
- **`Manage enrolments`**: The Instructor approves or drops individual student enrollments from the course management panel.
- **`Review Merge Proposal`**: Pre-condition: at least one student has submitted a Merge Proposal for this course. The Instructor inspects the proposed Course Material and either approves it (publishing it on the main course page with the contributor's name attached) or rejects it with an optional review note.

Student Use Cases

- **`Browse course materials`**: The student navigates to an enrolled course and opens individual sections to read the uploaded Course Materials.
- **`Clone course to Private Workspace`**: Pre-condition: the student is enrolled in a published course. The student triggers a clone action; the backend creates a Private Workspace containing mirror copies of all sections and Course Materials.
- **Submit Merge Proposal**: Pre-condition: the student owns a Private Workspace with at least one non-reference material or a modified reference material. The student selects materials and a target section, optionally adds a message, and submits the proposal for Instructor review.
- **`Post contextual message`**: The student types a message in the workspace chat, uses `@` to attach a Contextual Anchor to a specific Course Material, and submits. All enrolled users can see the message and follow the anchor.
- **`Query AI assistant`**: The student opens the AI panel, selects a Course Material for context, and types a question. The system sends the question and the extracted document text to the LLM API and displays the answer.

2.6 Existing Solutions

Several platforms address parts of the problem described in section 2.1. This section compares the most relevant ones against the core features that StudySpace provides, concluding with a synthesis of the integration gap that none of them fills end-to-end.

Table 2.1 Comparison of existing platforms against StudySpace features

Platform	Structured course materials	Student content contribution	Contextual discussion	AI content assistance	Role-based access control
Zoom / Teams	None	None	None	None	Partial
Discord	None	None	None	None	Partial
Moodle	Full	Limited	None	None	Full
Notion / Docs	Full	Partial	None	Generic	None
StudySpace	Full	Full	Full	Context	Full

The comparison in Table 2.1 shows that no existing platform covers all areas simultaneously. The following subsections analyse each platform in detail.

2.6.1 Zoom and Microsoft Teams

Zoom and Microsoft Teams are the dominant platforms for synchronous online education. Both support high-quality video conferencing, screen sharing, breakout rooms, and session recording. Microsoft Teams adds persistent channels, a file-sharing area backed by SharePoint, and deep integration with the Microsoft 365 productivity suite.

Despite these strengths, neither platform is designed for asynchronous, material-centric collaboration. Files shared during a meeting are stored in a flat SharePoint folder rather than being organised into a course structure that reflects the academic curriculum. Discussion threads are tied to meeting events, making conversations about a lecture slide inaccessible to students who join the course later. There is no mechanism for students to annotate shared files and propose changes back to the Instructor; the Instructor remains the sole publisher of content. The asynchronous knowledge-management and student contribution use cases are left entirely unaddressed.

2.6.2 Discord

Discord was originally designed for gaming communities but has been widely adopted by student groups as an informal alternative to institutional platforms. Its server and channel model allows students to self-organise into topic-specific channels, and its low barrier to entry means that most students already have an account.

However, Discord's design is fundamentally chat-centric and lacks any concept of structured educational content. Files are posted as raw message attachments in a general channel with no link to a course section or curriculum structure. There is no role hierarchy that distinguishes Instructors from students at the permission level beyond basic moderation roles; an Instructor cannot publish a set of Course Materials in a read-only area that students can then clone and annotate. Content disappears into scroll history, there is no access control aligned with course enrolment, and there is no mechanism for structured content contribution or AI-assisted querying of Course Materials.

2.6.3 Moodle

Moodle is a mature, open-source Learning Management System (LMS) used by thousands of universities worldwide, including CTU. It provides the most comprehensive course management feature set of any platform considered: Instructors can define courses with hierarchical sections, upload files and links, set up graded assignments and quizzes, track student progress, and manage enrolment through a role-based permission model.

Despite this breadth, Moodle has well-documented limitations with respect to student agency and contextual collaboration. Its discussion model is based on forums that are standalone modules attached to a course but not anchored to specific Course Materials; a student cannot attach a comment directly to a particular document without navigating to a separate forum module. Student contribution in Moodle is limited to wiki modules and assignment submissions, which are graded artefacts that do not become part of the course content visible to other students. Moodle also has no built-in AI assistant that operates on the actual course content in real time.

2.6.4 Notion and Google Docs

Notion and Google Docs represent a different category: general-purpose collaborative writing tools. Both support real-time co-editing, rich text formatting, embedded media, and comment threads. Notion extends this with a hierarchical page structure and customisable database views, making it possible to approximate a course workspace.

The critical gaps are the absence of role-based access control aligned with academic roles and the lack of a structured contribution workflow. In a shared Notion workspace, any collaborator can edit any page, creating conflicts and the risk of accidental data loss in a course setting with many students. Google Docs has stricter permission levels but no multi-tier workflow where a student creates a private draft, develops it independently, and then submits it for review and publication. Neither platform is AI-aware in a course-content sense: Notion AI and Google's Gemini integration can generate or summarise text, but they operate on the open document in isolation, without knowledge of which course the document belongs to or what other materials are in the same section.

2.6.5 Summary of Gaps

The analysis above reveals a consistent pattern across all evaluated platforms. None of them provides a mechanism for anchoring a discussion message to a named Course Material within a structured course hierarchy, meaning that contextual references must be described manually in text and the connection to the source material is easily lost. No platform supports a clone-then-propose contribution model in which a student creates a private enriched copy of a course section and submits it back for Instructor review and publication; the Instructor remains the sole publisher of course content in every platform examined. Where AI assistance is available, as in Notion AI or Google's Gemini integration, it operates on the open document in isolation and cannot be grounded in the broader course context. The result is that satisfying all four requirements simultaneously forces students to use at least two to three separate platforms, reproducing the context-switching overhead described in section 2.1.

StudySpace addresses this integration gap by combining structured course management, a clone-based student contribution workflow with Instructor review, chat messages with Contextual Anchors tied to specific Course Materials, and an AI assistant grounded in the actual course content, all within a single application with role-based access control.

2.7 Conclusions from Analysis

The analysis of the problem space, requirements, and existing platforms reveals a clear need for a unified system. None of the evaluated solutions successfully integrate structured course administration (FR-01) with a student-driven content extension system (FR-02) and contextual peer messaging (FR-03). Furthermore, the necessity for an AI assistant grounded directly in the course materials (FR-04) cannot be met by general-purpose collaborative tools. These identified gaps and defined requirements establish the boundaries and constraints that directly drive the architectural and technical design decisions presented in the following chapter.

Chapter 3

Design

Based on the requirements and gaps identified in Chapter 2, the following design decisions were made.

This chapter translates the requirements and architecture analysis established in Chapter 2 into a concrete technical design. The domain model defines the entities and their relationships. The system and logical architecture outlines the overarching application structure, while the database schema and entity-relationship diagram document the persistent data model. Finally, the API design and user interface design detail how the system will be accessed and experienced by its users.

3.1 Domain Model

The domain model consists of the entities that carry the application's persistent state. To ensure clarity, the complete model is divided into four functional subsystems corresponding to the core features described in Chapter 2. Each subsystem and its entities are described below. Additionally, the complete machine-readable domain model schema is provided in the enclosed electronic media as an attached resource.

3.1.1 User Management and Access Control

User is the central actor entity. It stores the account credentials (username, email, and bcrypt-hashed password), the display name, a profile picture URL, and a role field that takes one of two values: `STUDENT` or `INSTRUCTOR`. The entity also tracks a cumulative study-time counter and a current-streak counter that are updated by the gamification subsystem. A user can create many `Course` records (as an Instructor), own many `StudentWorkspace` records (as a Student), and participate in many `StudySession` records through the `SessionParticipant` junction entity.

Activity represents an event logged by the gamification subsystem, such as completing a study session. It carries a type, an optional text message, a timestamp, and a reference to the owning user, optionally linked to a specific study session.

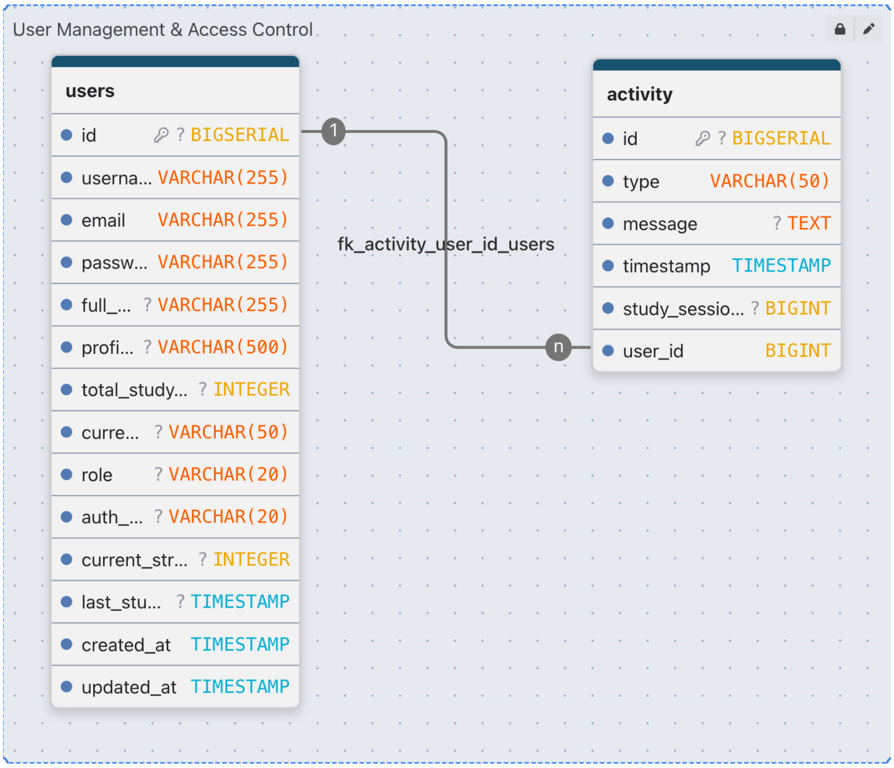


Figure 3.1 User management and access control domain model

3.1.2 Course Administration Module

Course represents a published course created by one Instructor. It carries a title, a text description, a boolean `isPublished` flag that controls student visibility, and a many-to-one relationship to its owning `User` (the Instructor). Deleting a `Course` cascades to all its `CourseSection` and `CourseEnrollment` records via `CascadeType.ALL` with orphan removal, ensuring no orphaned sections or enrollments remain after a course is removed.

CourseSection organises Course Materials within a course into named, ordered groups. It carries a title, an optional text description, and an `orderIndex` integer that determines the display sequence. Each section belongs to exactly one `Course`, and each section owns an ordered list of `CourseMaterial` records. Deleting a section cascades to all its materials.

CourseMaterial represents a single file uploaded into a course section. It stores a display title, the file storage URL, the inferred file type as a

`MaterialType` enum (`PDF`, `SLIDES`, `VIDEO`, `IMAGE`, `OTHER`), the original filename, an optional `contributorName` field (populated when the material originated from an approved Merge Proposal), and a timestamp. Each `CourseMaterial` belongs to exactly one `CourseSection`.

CourseEnrollment is the join entity that records a student's membership in a course. It carries an `EnrollmentStatus` field (`ACTIVE` or `DROPPED`) and the timestamp of the initial enrollment. A unique constraint on the pair `(course_id, student_id)` prevents duplicate enrollment rows. Setting the status to `DROPPED` rather than deleting the row preserves the enrollment history and allows re-activation.

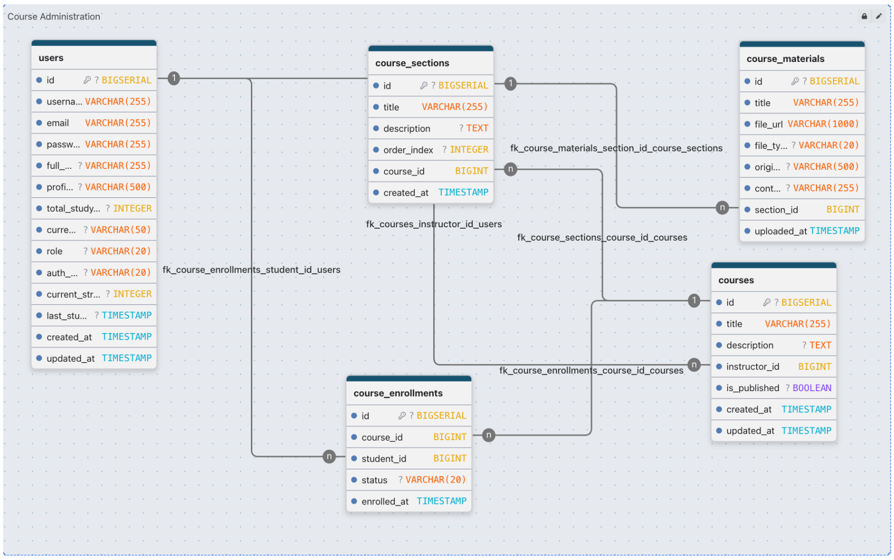


Figure 3.2 Course administration domain model

3.1.3 Content Extension System

StudentWorkspace is the root entity of a student's Private Workspace. It holds a name, a description, a reference to the owning `User`, and a list of `WorkspaceSpace` records. The workspace is the container that the student manages independently; it does not belong to any course and cannot be seen or modified by the Instructor.

WorkspaceSpace represents a single cloned or manually created space within a Private Workspace. It carries a title, a description, an optional foreign key to the `Course` it was forked from (`forkedFromCourse`), and a boolean `isPublished` flag used when a space is surfaced in a public gallery. Each space owns an ordered list of `WorkspaceSection` records.

WorkspaceMaterial mirrors `CourseMaterial` within the student's space. Two flags encode the copy-on-write behaviour: `isReference = true` means the material points to the original course file URL and no physical copy was made; `isHidden = true` implements a soft-delete that hides the row from the student's view without physically removing it, which is necessary because a contribution proposal may hold a foreign-key reference to the material row and deleting it would violate referential integrity.

ContributionProposal links a student's `WorkspaceMaterial` to a target `Course` and optionally a target `CourseSection`. Its `ProposalStatus` enum field (`PENDING`, `APPROVED`, `REJECTED`) drives the Instructor review workflow. When no existing section is targeted, the `proposedSectionTitle` field holds the name of a new section to be created on approval. The `reviewedAt` timestamp is set when the Instructor takes action, and the `contributorDisplayName` is copied from the student's profile at submission time so it remains stable even if the student later changes their name.

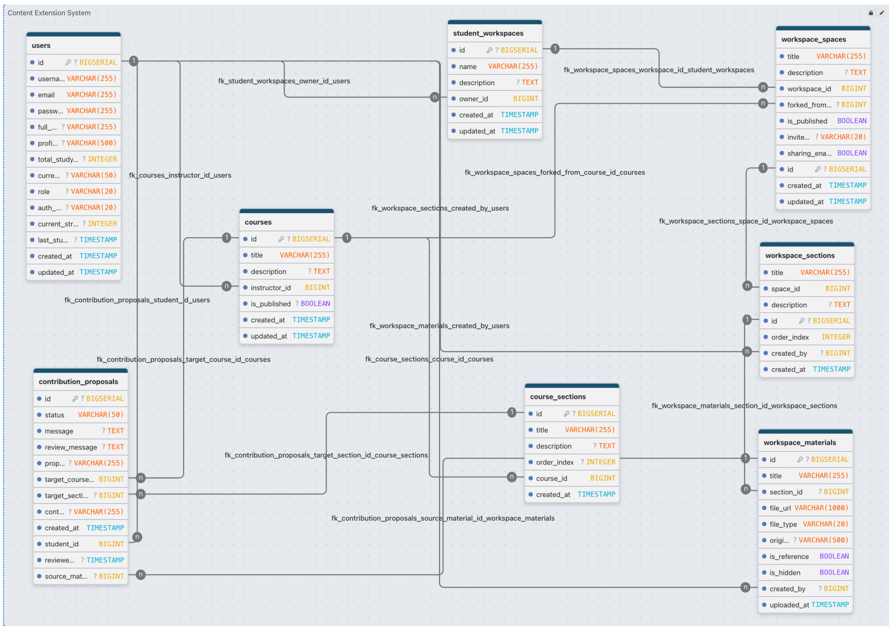


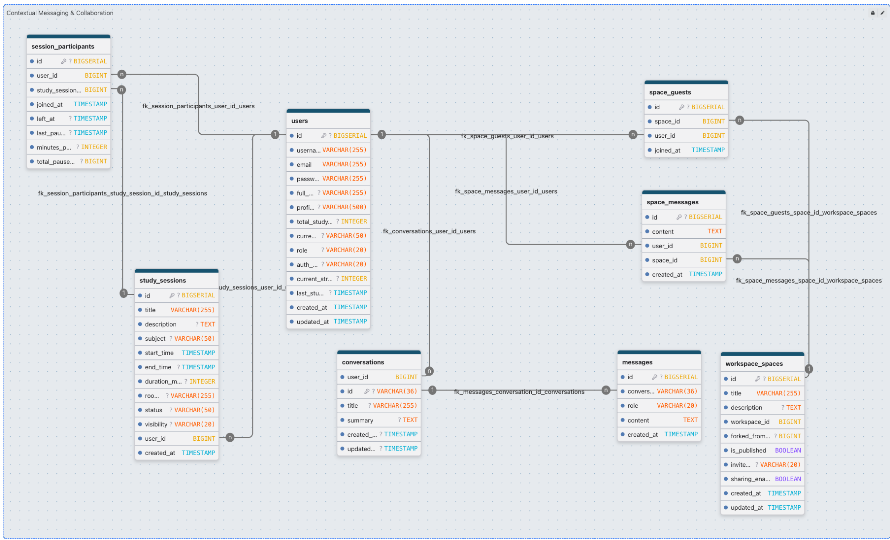
Figure 3.3 Content extension domain model

3.1.4 Contextual Messaging & Collaboration

StudySession and **SessionParticipant** manage real-time group study rooms, tracking the session lifecycle, visibility, and the cumulative study time of each participating student.

SpaceGuest and **SpaceMessage** enable real-time collaboration within a published Workspace Space, linking external users and their chat messages to the target space.

Conversation persists the AI chat memory for a single user session. Rather than storing every message individually in a single blob, the entity is part of a normalised schema tracking the user owner and the conversation title. Individual message turns are stored in the **Message** table. The conversation's primary key is a client-generated UUID string, so no database sequence is required.



■ **Figure 3.4** Contextual messaging and collaboration domain model

3.2 System and Logical Architecture

3.2.1 Conceptual Architecture

At the highest level, StudySpace is a web application that separates the browser-based user interface from the server-side application logic and the persistent data store. The user interface runs entirely in the browser and communicates with the backend over two channels: a standard request-response channel for operations such as loading course content, uploading materials, and submitting contribution proposals, and a persistent real-time channel for events that must reach all connected participants without a manual page refresh, such as chat messages and study-session activity feeds.

The backend is a single application that handles all business logic, enforces access control, processes file uploads, calls the external AI API, and broadcasts real-time events. The data store holds all persistent state, including user accounts, course structures, Private Workspace records, Merge Proposals, and conversation history.

All three tiers run as isolated containers managed by a single container orchestration file, so the complete application can be started with one command and requires no manual environment configuration. The final architecture diagram is presented in section **??** after the technology choices have been made.

3.2.2 System Architecture

StudySpace follows a three-tier architecture. The *presentation layer* is a Next.js and React application running in the browser; it communicates with the backend via the REST API for most operations and via a persistent WebSocket (STOMP/SockJS) connection during active study sessions. The *business logic layer* is a Spring Boot application that handles authentication, data access, real-time session broadcasting, and LLM API calls. The *data layer* is a PostgreSQL database that stores all persistent data.

For local development, all three tiers can be orchestrated using `docker-compose.yml` to run in the same virtual network. For production deployment, the architecture is decoupled: the frontend is deployed to a dedicated hosting platform (e.g., Vercel), the backend runs as a containerised service on a cloud platform (e.g., Render), and file storage is handled by an S3-compatible service (e.g., Tigris). This decoupling ensures independent scalability and simplifies deployment pipelines.

The backend is organised into the following packages, each with a single responsibility and dependencies flowing only from controller to service to repository:

Table 3.1 Backend package structure

Package	Responsibility
`controller`	Defines REST endpoints, maps HTTP requests to service calls, and validates incoming request bodies.
`service`	Contains all business logic. Services are called by controllers and may call one or more repositories. Transactions are managed at this layer.
`repository`	Spring Data JPA repositories, providing CRUD operations and custom JPQL queries for complex data access.
`entity`	JPA entity classes mapped to database tables.
`dto`	Data Transfer Objects used as the API contract, decoupling the API response shape from the internal entity structure.
`security`	JWT filter, `UserDetailsService` implementation, and Spring Security configuration.
`config`	Application-level configuration beans: CORS policy, Swagger setup, and WebSocket registration.

3.2.3 Logical Architecture

The server-side application follows the Model-View-Controller (MVC) layered pattern. The *controller layer* receives HTTP requests, delegates to the appropriate service, and serialises the response. The *service layer* contains all business logic, orchestrates data access, and enforces domain rules such as ownership checks and status transitions. The *repository layer* provides data access through named query methods and custom queries for complex lookups. Entity classes map the relational schema to Java objects. Data Transfer Objects (DTOs) define the API contract and decouple the internal entity structure from the shape of the JSON responses, so the two can evolve independently.

3.3 File Storage

FR-01 and FR-02 require file upload and retrieval for Course Materials and Private Workspace materials. The storage backend must be swappable between a local development mode and a production cloud target.

Local file storage (writing files to a directory under the working directory) is zero-cost and requires no external service in development. It is unsuitable for production because files are lost when the container is recreated and cannot be shared across multiple server instances.

Google Cloud Storage (GCS) is a managed object store with high durability, direct URL serving, and a Java client library. It requires a GCP project and credentials, which adds setup overhead during development.

A `FileStorageService` interface with two concrete implementations was designed: `LocalFileStorageService` for development and `GcsFileStorageService` for production. The active implementation is selected by a Spring profile, keeping the storage backend transparent to the rest of the application and satisfying NFR-04. The interface defines three methods: `store`, `copy`, and `delete`, which are the only storage operations needed by FR-01 and FR-02.

3.4 LLM API Integration

FR-04 requires an LLM API capable of answering questions grounded in a provided document context within a single prompt, without fine-tuning.

Google Gemini API (gemini-3-flash-preview) provides a large context window, competitive quality on open-domain question answering, and an official Java SDK that integrates directly with the Spring Boot application. The free tier is sufficient for prototype-level usage.

OpenAI GPT-5.4-mini exposes a compatible REST interface and achieves comparable quality on question answering tasks. However, the Java SDK is less mature than the official Google GenAI SDK, and the cost per token is higher for the same context window size at the time of writing.

The Google Gemini API was selected as the default provider because the official Google GenAI Java SDK [14] integrates directly with Spring Boot without additional HTTP client configuration, the `gemini-3-flash-preview` model's context window comfortably accommodates the semantically relevant chunks retrieved from the `pgvector` database required by FR-04, and the free tier capacity is sufficient for the prototype.

3.5 Database Schema

The relational schema follows directly from the domain model described in section 3.1. Each entity maps to a dedicated table, and all foreign-key relationships are enforced at the database level.

The primary tables correspond to the core domain entities: users, courses, course enrollments, course sections and materials, and their workspace counterparts. The `course_enrollments` table enforces a unique constraint on the column pair `(course_id, student_id)` to prevent duplicate enrollment rows. The `workspace_materials` table carries the two boolean flags `isReference` and `isHidden` that implement the copy-on-write and soft-delete semantics described in section 3.1.

The AI conversation memory utilises two tables: `conversations` stores the overarching context and the rolling `summary` of compressed long-term memory, while the `messages` table records individual chat interactions. This separation allows efficient querying of recent conversation history. For document-grounded question answering (FR-04), the system uses the `document_chunks` table. This table leverages the `pgvector` PostgreSQL extension to store high-dimensional embeddings of document segments, enabling fast semantic similarity search during the retrieval-augmented generation process.

3.6 API Design

The backend exposes a RESTful API following standard HTTP conventions. Resources are identified by URL paths, and operations are expressed through HTTP methods (`GET`, `POST`, `PUT`, `DELETE`). All protected endpoints require a valid JWT token in the `Authorization: Bearer <token>` header.

Authentication follows a four-step flow. The client sends credentials to `POST /api/auth/login`. The server validates the credentials and returns a signed JWT token. The client stores the token in memory and includes it in the `Authorization` header of all subsequent requests. The server's `JwtAuthenticationFilter` validates the token on each request before passing control to the controller.

The full API is documented through the Swagger UI available at `/swagger-ui/index.html` when the backend is running, auto-generated by Springdoc OpenAPI from controller annotations.

3.7 UI and Wireframe Design

The interface is role-aware: students and Instructors see different navigation options and action buttons depending on their role. The application uses a sidebar navigation layout with the main content area to the right.

3.7.1 Navigation and Layout

The sidebar items differ by role. Students see links to their enrolled courses, their Private Workspaces, and their study sessions. Instructors additionally see course management, enrollment management, and Merge Proposal review links. The role check is performed client-side from the JWT payload, but all data-sensitive actions are re-validated server-side.

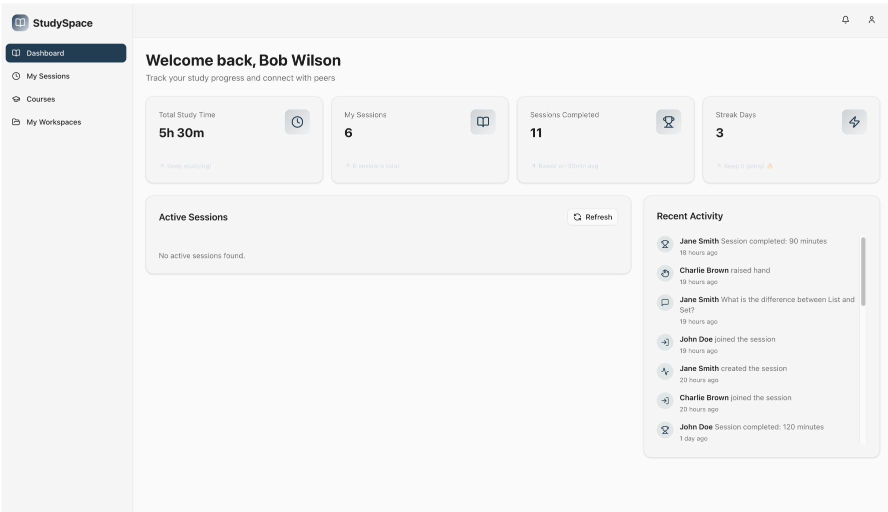


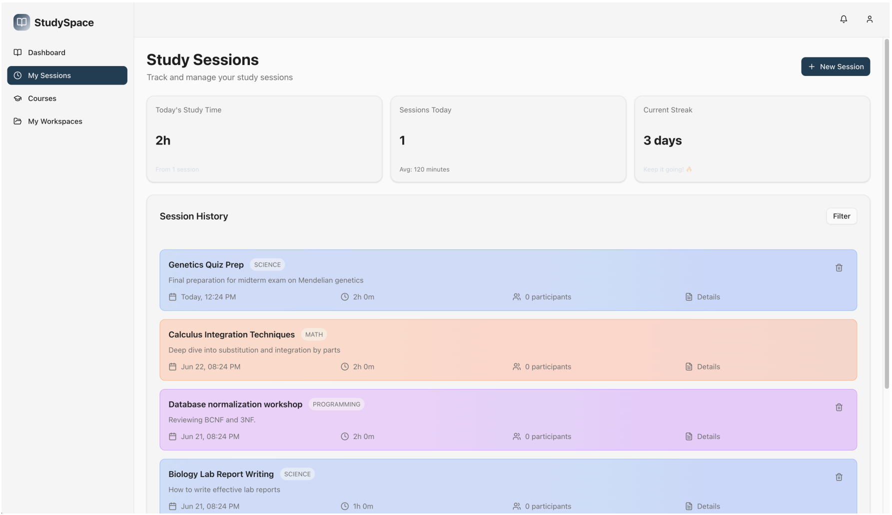
Figure 3.5 Student dashboard: The primary entry point featuring recent activities and quick navigation

3.7.2 Course and Workspace Views

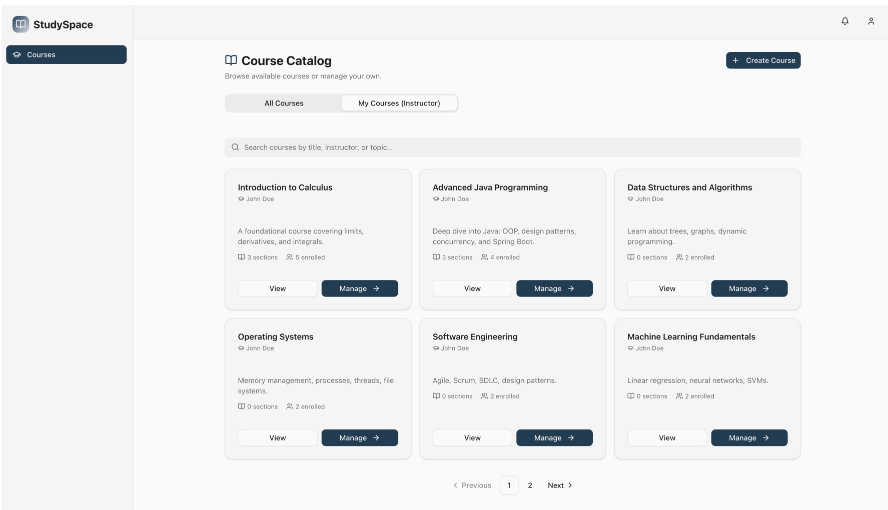
The course list page shows all courses the user is enrolled in or manages.

Selecting a workspace opens the workspace viewer, which renders the Course Material alongside a contextual chat panel. Students can use the chat panel to discuss the course content and attach a Contextual Anchor to a specific Course Material using the `@`-mention system.

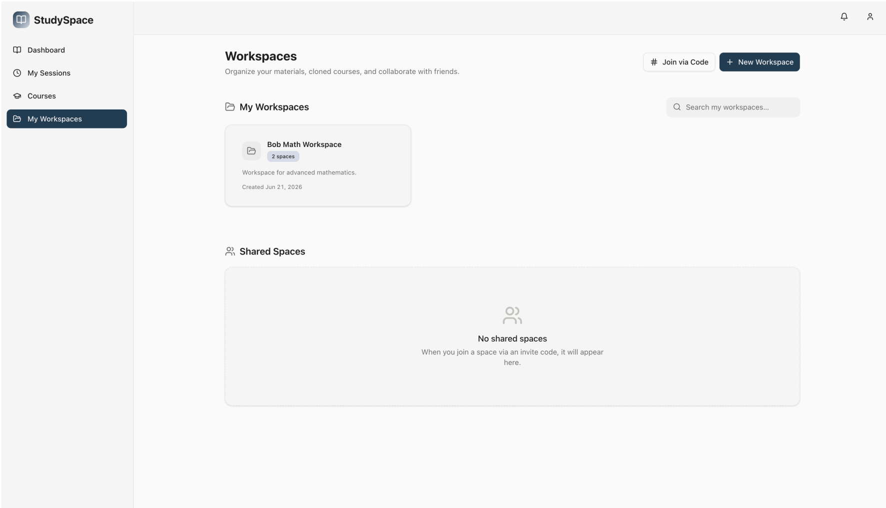
The workspace viewer places the material list on the left and the contextual chat panel on the right. This layout satisfies the FR-03 use case `Post contextual message` by keeping the source material and the discussion visible simultaneously. The panel is resizable so students can balance reading space against chat space according to their preference.



■ **Figure 3.6** Study sessions view: Managing active and scheduled collaborative sessions



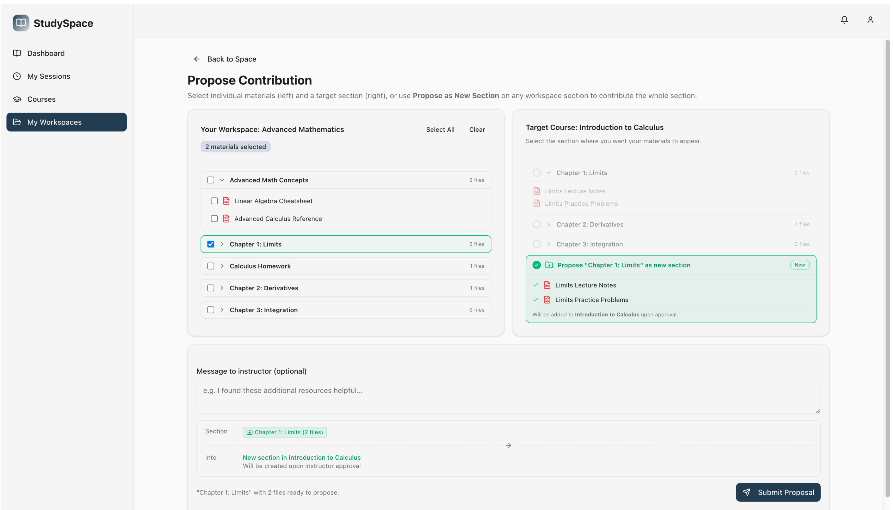
■ **Figure 3.7** Instructor courses view: Managing active course structures and enrollments



■ **Figure 3.8** Workspace viewer: Course Material list with contextual chat panel

3.7.3 Private Workspace and Merge Proposal Views

When a student clones a course, they receive a Private Workspace that they can edit freely. The cloned workspace view mirrors the course structure and adds upload controls for new materials. The Merge Proposal submission dialog allows the student to select materials and choose a target section, optionally typing a proposed new section name if none of the existing sections is appropriate.



■ **Figure 3.9** Merge Proposal submission dialog

The Merge Proposal dialog reflects the FR-02 use case `Submit Merge Proposal`: the student selects one or more materials, picks a target section or proposes a new one, and submits. The design keeps the form minimal to reduce friction and mirrors the Git-inspired workflow that technically experienced students will recognise.

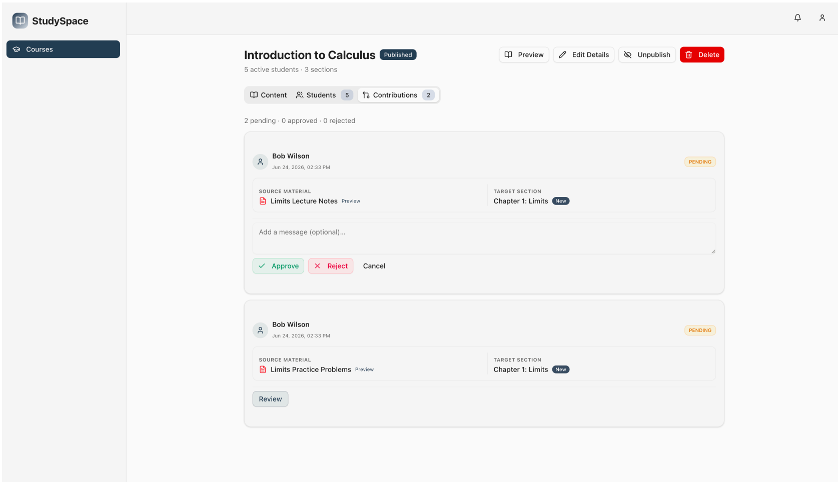


Figure 3.10 Instructor review view: Evaluating a student's merge proposal

3.7.4 AI Assistant Panel

The AI assistant is accessible through a collapsible side panel within the workspace viewer. The student types a question in natural language, and the system sends the question together with the content of the currently open Course Material to the LLM API. The response is displayed in a conversational format with a typewriter animation. The panel is designed not to obscure the Course Material so the student can consult both simultaneously.

The collapsible design satisfies FR-04 by making the AI assistant available at any point during material review without forcing the student to navigate to a separate page, preserving the reading context while the answer is displayed.

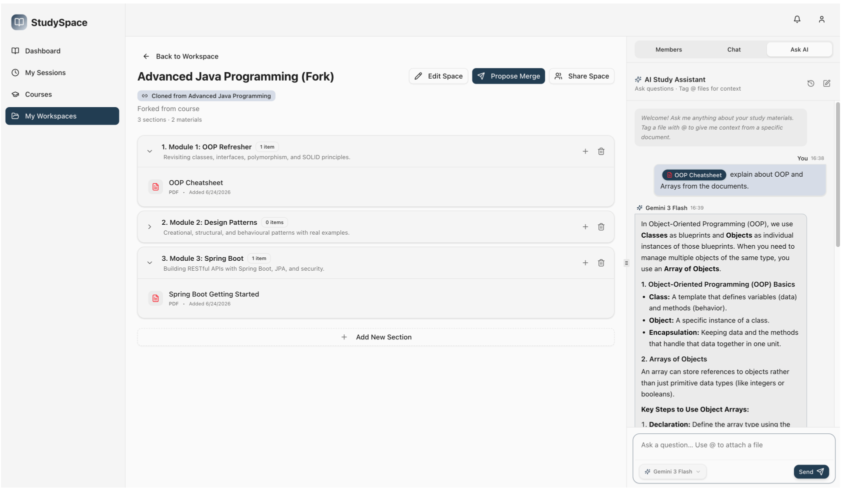


Figure 3.11 AI assistant panel: Conversational interface for querying course materials

Chapter 4

Implementation

This chapter documents how the five core features defined in Chapter 2 were translated into working code. Each feature section follows the same structure: the purpose and technical challenge are contextualised first, the application screenshot shows the feature in action, the underlying logic is explained in prose, and a targeted code listing illustrates the most important implementation decision. Section 4.6 covers automated and usability testing, and section 4.7 describes the project organisation.

4.1 Course Administration Module (FR-01)

The course administration feature addresses FR-01: Instructors must be able to create and manage courses, upload Course Materials into named sections, and control student enrollment. The technical challenge in this feature is keeping the file storage backend transparent to the business logic so that the development environment (local disk) and the production environment (Google Cloud Storage) can be swapped by changing a Spring profile, without any changes to the service layer.

The course management page presents the Instructor with a list of sections and a file upload button per section. Selecting a file triggers a multipart form data request to the backend; the Instructor does not see storage paths or file types, which are resolved automatically by the backend.

The `CourseService` delegates all file operations to the `FileStorageService` interface. When `uploadMaterial` is called, the service stores the file, infers the `MaterialType` from the file extension, and persists the resulting URL in the `CourseMaterial` entity. The ownership check in `assertInstructor` ensures that only the course owner can modify sections or materials; any other authenticated user receives an HTTP 403 response. The listing below shows the upload method, which encapsulates this flow in under twenty lines.

```
public CourseMaterialDTO uploadMaterial(Long sectionId, Long
    requestingUserId,
                                              MultipartFile file, String
    title) {
    CourseSection section = findSection(sectionId);
    // Only the course instructor may upload
    assertInstructor(section.getCourse(), requestingUserId);

    // FileStorageService is injected; impl selected by Spring
    profile
    String fileUrl = fileStorageService.store(file, "courses");
    MaterialType fileType = detectFileType(file.getOriginalFilename()
    );

    CourseMaterial material = CourseMaterial.builder()
            .title(title).fileUrl(fileUrl).fileType(fileType)
            .originalFileName(file.getOriginalFilename())
            .section(section).build();
    return toMaterialDTO(materialRepository.save(material));
}
```

Code listing 4.1 Code listing 4.1 — Course material upload with storage abstraction (CourseService.java)

The `FileStorageService` interface defines three methods (`store`, `copy`, `delete`), and the active implementation is injected by Spring. This design satisfies NFR-04 (maintainability) by keeping the service layer unaware of the storage backend, and means that switching to a cloud provider for production requires only an environment variable change, not a code change.

4.2 Content Extension System (FR-02)

The Content Extension System addresses FR-02: students must be able to clone a course into a Private Workspace, modify it freely, and submit a Merge Proposal for the Instructor to review. The key technical challenges are the copy-on-write cloning strategy (avoiding redundant file copies for shared base materials) and the referential-integrity constraint that prevents hard deletion of materials that are still referenced by pending proposals.

The workspace view shows the student a mirror of the course structure. An upload button per section allows the student to add new materials. The Merge Proposal dialog appears when the student selects one or more materials and clicks the contribute button; the student picks a target section (or names a new one) and submits.

When `forkCourse` is called on `WorkspaceService`, the backend iterates over all sections and Course Materials of the source course and creates `WorkspaceSection` and `WorkspaceMaterial` rows that point to the original file URLs (`isReference = true`) rather than copying the files. Only materials uploaded by the student later are stored as new files (`isReference = false`). When a student removes a reference material from their view, `deleteMaterial` in `WorkspaceService` checks for active `ContributionProposals`. If a proposal exists, it sets `isHidden = true` on the row instead of deleting it to preserve referential integrity; otherwise, it deletes the reference cleanly. The listing below shows this logic.

```java
public void deleteMaterial(Long materialId, Long userId) {
    WorkspaceMaterial material = materialRepository.findById(materialId)
            .orElseThrow(() -> new RuntimeException("Material not found: " + materialId));

    // Authorisation checks omitted for brevity

    boolean hasProposal = proposalRepository.existsBySourceMaterialId(materialId);
    if (hasProposal) {
        // Soft-delete: keep the row so ContributionProposal FK stays intact
        material.setIsHidden(true);
        materialRepository.save(material);
    } else {
        // Safe to hard-delete
        if (!Boolean.TRUE.equals(material.getIsReference())) {
            fileStorageService.delete(material.getFileUrl());
        }
        materialRepository.delete(material);
    }
}
```

Code listing 4.2 Code listing 4.2 — Delete strategy for reference materials (WorkspaceService.java)

The two-branch delete strategy preserves referential integrity without requiring a separate audit table or a deferred constraint. The `WorkspaceSection DTO` mapper already filters hidden materials before returning the section content to the frontend, so the student's view is consistent with the logical deletion even though the database row remains.

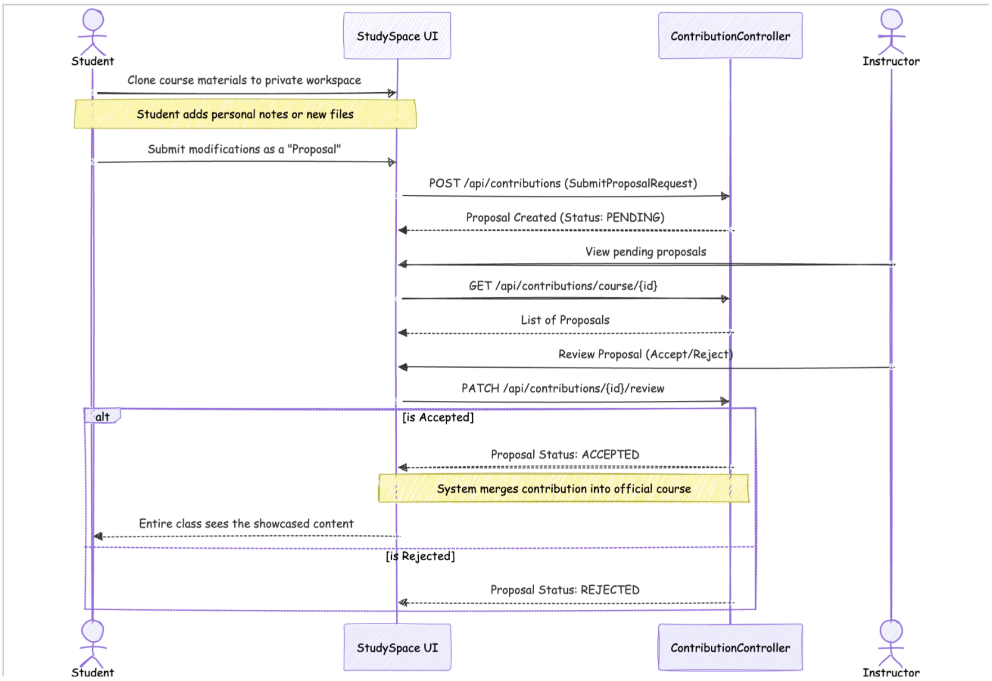


Figure 4.1 Merge proposal lifecycle sequence diagram: From student submission to instructor review and acceptance

4.3 Contextual Messaging System (FR-03)

The Contextual Messaging System addresses FR-03: users must be able to post messages in the workspace chat and attach a Contextual Anchor to a specific Course Material so that other users can open the material from within the message. The technical challenge is embedding a non-editable interactive chip into a plain-text-like input field so that the anchor is preserved through submission, serialisation, and re-render.

The chat panel is shown after a student typed @lecture and the autocomplete dropdown appeared. After selecting a Course Material, a styled chip replaces the @-query in the input. The submitted message is rendered with a clickable badge that opens the file.

The message input uses a <div contentEditable> element rather than a <textarea> because <textarea> can only hold plain text, whereas a contentEditable div can host arbitrary DOM nodes. When the user selects a Course Material from the autocomplete, the insertTag function deletes the @<query> text and inserts a <span> element with contentEditable="false" and data-materialId, data-materialTitle, and data-materialUrl attributes. Before submission, parseInput serialises the span into the token @[<id>:<title>], which is stored as the message text. On render, a regular expression parses the token back into a <Badge> component. The listing below shows the serialisation step.

```
// Walks the contentEditable div's child nodes and builds the token
    string
const parseInput = (): string => {
  if (!inputRef.current) return "";
  let question = "";
  for (const node of Array.from(inputRef.current.childNodes)) {
    if (node.nodeType === Node.TEXT_NODE) {
      question += node.textContent;
    } else if (node.nodeType === Node.ELEMENT_NODE) {
      const el = node as HTMLElement;
      if (el.dataset.materialId) {
        // Non-editable chip -> portable anchor token
        question += `@[${el.dataset.materialId}:${el.dataset.
    materialTitle}]`;
      } else {
        question += el.textContent ?? "";
      }
    }
  }
  return question.trim();
};
```

Code listing 4.3 Code listing 4.3 — DOM serialisation to portable anchor token (member-chat.tsx)

The token format `@[id:title]` is compact, survives JSON serialisation, and is unambiguous to parse with a single regular expression on the render side. Storing the material ID inside the token allows the backend to validate or re-fetch the material metadata if needed, while the title provides a human-readable fallback if the material is later renamed.

4.3.1 Workspace Collaboration

To facilitate peer-to-peer discussion, Private Workspaces must support collaboration among multiple students. The system achieves this by allowing the workspace owner to add other users as members. The `WorkspaceService` manages this by persisting a list of `SpaceMemberDTO` objects associated with the space. Before a user can post or view messages in the chat, the `assertMember` security check verifies their membership. This transforms the Private Workspace from an isolated clone into a collaborative study environment, directly supporting the interactive requirements of FR-03.

4.3.2 Real-Time Communication via WebSockets

For the discussion to be truly interactive, messages must be delivered to all workspace members instantly without client-side polling. This is implemented using WebSockets with STOMP over SockJS (as justified in section 2.2.5). While earlier iterations used WebSockets exclusively for study sessions, the infrastructure was extended to power the Contextual Messaging System.

When a user submits a message, the `SpaceChatController` intercepts the HTTP `POST` request, saves the message to the database, and immediately broadcasts the message payload to the STOMP destination corresponding to the workspace ID.

```java
@PostMapping
public ResponseEntity<SpaceMessageDTO> sendMessage(
        @PathVariable Long spaceId,
        @RequestBody Map<String, Object> payload) {

    // Parse payload and save SpaceMessage entity...
    SpaceMessageDTO dto = SpaceMessageDTO.builder()
            .content(message.getContent())
            .spaceId(message.getSpace().getId())
            // ... other fields omitted for brevity
            .build();

    // Broadcast to space members over WebSocket channel
    String destination = "/topic/space/" + spaceId + "/chat";
    messagingTemplate.convertAndSend(destination, dto);

    return ResponseEntity.status(HttpStatus.CREATED).body(dto);
}
```

Code listing 4.4 Code listing 4.4 — WebSocket broadcast for workspace chat (SpaceChatController.java)

On the frontend, the React application uses a WebSocket hook to subscribe to the `/topic/space/{spaceId}/chat` channel. Upon receiving a message event, the client appends the new message to the local component state, triggering an immediate UI update for all connected members.

4.4 AI-Based Content Querying (FR-04)

The AI querying feature addresses FR-04: users must be able to ask natural-language questions about the content of the currently open workspace and receive answers grounded in that content. The technical challenges are normalising three distinct document URL formats into a single byte stream, constructing a structured prompt that keeps the model grounded in the document, and handling API rate limits gracefully so the student always receives a useful response.

The AI panel shows the student's question at the top and the LLM-generated answer below, displayed with a typewriter animation. The panel is collapsible so the student can read the Course Material alongside the answer.

When the user submits a question, the frontend sends the question and the selected workspace references to `POST /api/chat/query`. The `DocumentVectorService` retrieves semantically relevant document excerpts from the `pgvector` database by comparing the vector embedding of the user's question against the stored document chunks (which are extracted via Apache PDFBox [15] and embedded during file upload). The `PromptBuilder` then constructs a comprehensive prompt consisting of a system preamble, conversation history, the retrieved RAG excerpts, and the user's question. Finally, the configured `LlmProvider` calls the underlying API (e.g., via the Google GenAI Java SDK [14]). The listing below shows the structured prompt assembly.

```java
public String buildRuntimePrompt(String summary, List<Message> recentMessages,
                                 List<String> ragChunks, String userQuestion) {
    StringBuilder sb = new StringBuilder();
    sb.append("You are a teaching assistant. Prefer the provided document " +
              "context over your own knowledge when it is relevant.\n\n");

    if (summary != null && !summary.isBlank()) {
        sb.append("[Previous context]\n").append(summary).append("\n\n");
    }
    if (ragChunks != null && !ragChunks.isEmpty()) {
        sb.append("[Document Excerpts]\n");
        ragChunks.forEach(chunk -> sb.append(chunk).append("\n---\n"));
    }
    sb.append("[Current Question]\n").append(userQuestion);
    return sb.toString();
}
```

Code listing 4.5 Code listing 4.5 — Structured prompt construction (PromptBuilder.java)

The system instruction instructs the model to prefer the document context, which is the core of the Retrieval-Augmented Generation (RAG) pattern implemented here. Truncating the context at 10 000 characters controls per-request cost while keeping the most relevant part of the document available to the model. When the API returns a `429 RESOURCE_EXHAUSTED` status, the service parses the retry delay and surfaces it to the user as a human-readable message rather than propagating a raw exception.

4.5 Access Control (FR-05)

The access control feature addresses FR-05: the system must distinguish Instructors from Students at the API level and reject unauthorised requests before they reach the service layer. The technical challenge is making the authentication filter stateless so that no server-side session store is required, while still providing the full user identity (including role) to every downstream service call.

The two navigation sidebars illustrate the role distinction. The student sees course browsing and workspace links; the Instructor additionally sees the course management and Merge Proposal review links. Both views are generated from the same component by reading the role field from the JWT stored in the React context.

When a user authenticates, `AuthService` validates the credentials, looks up the `User` entity, and calls `JwtUtil.generateToken` to produce a signed JWT embedding the user's email as the subject. The token is signed with HMAC-SHA256 and expires after ten hours. On every subsequent request, `JwtAuthenticationFilter` extracts the token from the `Authorization:Bearer` header, validates the signature and expiry, loads the `UserDetails` by email, and populates the Spring Security context. The listing below shows the core of the filter.

```java
// Runs once per request; short-circuits if no token is present
String authHeader = request.getHeader("Authorization");
String jwt = null;

if (authHeader != null && authHeader.startsWith("Bearer ")) {
    jwt = authHeader.substring(7); // strip "Bearer " prefix
} else if (request.getParameter("token") != null) {
    jwt = request.getParameter("token"); // WebSocket fallback
}

if (jwt == null) { filterChain.doFilter(request, response); return; }

String userEmail = jwtUtil.extractUsername(jwt); // decode subject claim
if (userEmail != null
        && SecurityContextHolder.getContext().getAuthentication() == null) {
    UserDetails userDetails = userDetailsService.loadUserByUsername(userEmail);
    if (jwtUtil.isTokenValid(jwt, userDetails)) {
        // Populate security context so @PreAuthorize annotations work
        UsernamePasswordAuthenticationToken authToken =
            new UsernamePasswordAuthenticationToken(
                userDetails, null, userDetails.getAuthorities());
```

```
            SecurityContextHolder.getContext().setAuthentication(authToken);
        }
    }
    filterChain.doFilter(request, response);
```

Code listing 4.6 Code listing 4.6 — JWT validation in the authentication filter (JwtAuthenticationFilter.java)

The filter populates the Spring Security context with the user's authorities (which encode the role), enabling `@PreAuthorize` annotations and explicit `assertInstructor` checks in the service layer to reject Student requests on Instructor-only operations. The stateless approach means that the server holds no session state between requests, satisfying NFR-02 and simplifying horizontal scaling.

4.6 Testing

4.6.1 Backend Automated Tests

The backend uses three levels of automated tests, all implemented with JUnit 5 and the Spring Boot Test framework.

Unit tests cover individual service classes in isolation. Dependencies are substituted with Mockito mocks so that each test exercises exactly one unit. Examples include `UserServiceTest`, `StudyGroupServiceTest`, and `GamificationServiceTest`.

Repository tests verify that the Spring Data JPA repositories produce correct SQL and return expected entity graphs. They run against an in-process H2 database seeded with test data. `UserRepositoryTest` checks that user look-up by email and username behaves correctly.

Integration tests exercise complete request-response cycles through the Spring MVC layer. `UserControllerIntegrationTest` and `StudySessionControllerIntegrationTest` spin up a `@SpringBootTest` application context, issue HTTP requests via `MockMvc`, and assert on the HTTP status code and the JSON response body. `GlobalExceptionHandlerTest` verifies that the centralised exception handler maps domain exceptions to the correct HTTP status codes.

During development, several important bugs were found and corrected through testing. A referential-integrity violation when deleting reference materials was resolved by introducing the soft-delete `isHidden` flag described in section 4.2. A scroll-hijacking issue in the AI chat panel was discovered during manual end-to-end testing and fixed by introducing a `userScrolledRef` guard that suppresses programmatic scroll when the user has manually scrolled upward.

4.6.2 Frontend Automated Tests

Complementing the backend testing suite, the frontend employs a dual testing framework strategy to verify user interfaces and application flows.

Unit and integration tests are executed using Vitest and the React Testing Library. Components such as the authentication forms are mounted in an isolated `jsdom` environment. By mocking the Next.js router and authentication contexts, the tests simulate user input and assert on the resulting DOM updates and validation errors without relying on a running server.

End-to-end (E2E) tests are orchestrated with Playwright. These tests spin up a headless browser (such as Chromium) and navigate through complete user journeys. By leveraging Playwright's network interception capabilities, the tests mock the backend API responses. This approach ensures that the frontend behaves correctly under various backend states—such as a successful login or an HTTP 401 Unauthorized response—in a highly deterministic and fast-executing manner.

4.6.3 User and Usability Tests

Each test below is written as a numbered step sequence. Pre-conditions and edge cases are noted inline.

Test 4.1 — Upload and browse a Course Material (FR-01)

1. Log in as an Instructor account.
2. Navigate to an existing published course and open the course management view.
3. Select a section and click the upload button. Choose a PDF file of at least 1 MB.
4. Confirm the upload. Expected: the material appears in the section list with the correct file name and a PDF icon.
5. Log out. Log in as an enrolled Student. Navigate to the same course and section.
6. Confirm the material is visible and can be downloaded.

Edge case: Attempt the upload step (step 3) while logged in as a Student. Expected: the upload button is absent from the UI; a direct API call to `POST /api/courses/sections/id/materials` returns HTTP 403 Forbidden.

Test 4.2 — Clone course and submit Merge Proposal (FR-02)

1. Log in as a Student enrolled in a published course.

2. Navigate to the course and click the “Clone to workspace” button.
3. Confirm the Private Workspace is created with sections and materials mirroring the original course.
4. Upload a new PDF into one of the workspace sections.
5. Select the uploaded material, click “Contribute”, choose an existing target section, optionally add a message, and submit the Merge Proposal.
6. Log out and log in as the course Instructor. Navigate to the Merge Proposal review tab.
7. Confirm the proposal appears with status `PENDING`. Approve it.
8. Navigate to the course section. Confirm the approved material appears with the contributor’s name.

Edge case: In step 5, select a reference material (one cloned from the course) rather than a newly uploaded file. Expected: the proposal is accepted, but the backend copies the file from the original URL into the course storage, not from a non-existent student upload.

Test 4.3 — Post a message with Contextual Anchor (FR-03)

1. Log in as a Student and open a workspace.
2. In the chat panel, type `@` followed by part of a Course Material name.
3. Confirm the autocomplete dropdown appears with matching materials.
4. Select a material. Confirm the chip is inserted inline in the message input.
5. Submit the message. Confirm the chip is rendered as a clickable badge in the chat.
6. Click the badge. Confirm the Course Material file is downloaded or opened.

Edge case: Open the same workspace as a second enrolled Student. Confirm the message with the Contextual Anchor is visible to the second student and the badge is functional.

Test 4.4 — Query AI assistant (FR-04)

1. Log in as a Student, open a workspace, and select a Course Material (PDF).
2. Open the AI assistant panel and type a factual question about the document’s content.
3. Submit. Confirm the response appears with a typewriter animation and references concepts from the document rather than generic knowledge.

4. Type a follow-up question. Confirm the AI response reflects the conversation history.

Edge case: Open the AI panel without selecting any Course Material and submit a question. Expected: the backend sends the question without document context and the AI responds with general knowledge; no error is shown.

Test 4.5 — Role-based access control (FR-05)

1. Log in as a Student. Open the browser developer tools and copy the JWT from the `Authorization` header of any API request.

2. Using `curl` or Swagger UI, call `POST /api/courses` (Instructor-only endpoint) with the Student JWT.

3. Expected: the server returns HTTP 403 Forbidden with an error body.

4. Repeat for `PUT /api/courses/{id}/sections` and `POST /api/proposals/{id}/review`.

5. In each case confirm the response is 403 and the operation has no effect.

Edge case: Submit a request with a malformed or expired JWT to any protected endpoint. Expected: the server returns HTTP 401 Unauthorized with the message "Unauthorized".

4.7 Project Organisation

4.7.1 Workflow and Version Management

The project is maintained in a single Git repository with a monorepo layout separating the `backend/`, `frontend/`, and `documentation/` directories. Development follows a branch-based workflow: each feature or fix is developed on a dedicated branch and merged into `main` via a pull request after passing all automated checks. Commits follow the Conventional Commits specification to keep the history readable and to support automated changelog generation.

4.7.2 CI/CD Pipeline

A GitLab CI/CD pipeline runs automatically on every push on master branch. The pipeline defines multiple stages to guarantee code quality and automate deployment. The first stages build the backend with Gradle and the frontend with `npm`, followed by comprehensive unit, integration, and end-to-end (Playwright) testing. Subsequent stages run security and code quality scans, including OWASP Dependency-Check and SonarQube. If all checks pass on the main branch, the final stage automatically deploys the frontend to Vercel.

A separate pipeline hook triggers the backend container deployment to the Render cloud platform.

The pipeline enforces that `master` is always in a deployable state. No code reaches the branch without passing the backend tests and the frontend type check, which together cover the most common categories of regression.

4.7.3 Documentation

The thesis source is maintained as a LaTeX project using the FIT CTU thesis template. The `ctufit-thesis.tex` root file includes the chapter files from the `text/` subdirectory. Figures are stored under `documentation/images/` and referenced from the chapter files. The bibliography is maintained in `text/bib-database.bib` and compiled with Biber.

The backend API is documented automatically by Springdoc OpenAPI, which generates a Swagger UI from the controller annotations and makes it available at `/swagger-ui/index.html` when the application is running. Public service methods carry JavaDoc comments that explain the purpose and contract of each operation.

Chapter 5
Conclusion

StudySpace was designed to address the tool fragmentation that forces students and instructors at higher-education faculties to operate across at least two or three disconnected platforms to accomplish tasks that are intrinsically linked: reading a document, asking a question about it, and contributing an improvement to it. The project demonstrates that a single, integrated environment can satisfy all four core collaboration needs simultaneously.

The course administration module (FR-01) was fully implemented as a hierarchical structure of courses, sections, and Course Materials with Instructor-controlled enrollment and a pluggable file storage backend that switches between local disk and Google Cloud Storage by changing a Spring profile. The Content Extension System (FR-02) realised the clone-then-propose workflow: students can fork any enrolled course into a Private Workspace, extend it with their own materials, and submit a Merge Proposal for the Instructor to review and publish, with the contributor's name recorded on the resulting Course Material. The Contextual Messaging System (FR-03) introduced a portable anchor token that survives serialisation and re-render, enabling any message in the workspace chat to carry a deep link to a specific Course Material that other students can click to open the file directly. The AI content querying feature (FR-04) integrated the Google Gemini API through a structured prompt pipeline that grounds the model's answers in the actual course document rather than in general knowledge, with a hybrid memory model that preserves conversation context across long sessions. The access control layer (FR-05) enforces role separation at the API level through a stateless JWT filter, ensuring that Student accounts cannot execute Instructor-only operations regardless of the client application used.

User and usability testing confirmed the correctness of the main flows and uncovered two bugs that were resolved during the implementation phase: a referential-integrity violation when removing reference materials, addressed by the soft-delete `isHidden` flag, and a scroll-hijacking issue in the AI panel, resolved by a programmatic scroll guard.

5.1 Future Work

The current prototype has several documented limitations that were explicitly deferred to keep the scope manageable for a bachelor thesis.

The Contextual Anchor in the messaging system (FR-03) currently stores only the material identifier and title in the token. A natural extension is to support page-level or section-level anchors, where the token also encodes a byte offset or a page number, allowing the receiving client to open the PDF viewer at the exact page the sender was looking at.

The enrollment model (FR-01) supports two statuses (`ACTIVE` and `DROPPED`) but does not model enrollment requests or an Instructor approval step. In a real faculty deployment, courses that should not be publicly accessible would require an invitation or approval workflow before granting access to Course Materials.

The study session feature, implemented through the WebSocket infrastructure, provides real-time participant tracking and a shared activity feed. Based on the testing conducted in section 4.6, a shared collaborative document or whiteboard within the session would significantly increase the usefulness of synchronous study sessions.

Finally, the application was developed with a containerised local environment and deployed to modern PaaS platforms (Vercel and Render). While these platforms handle initial scaling automatically, making the backend truly horizontally scalable under heavy load would require the WebSocket session state to be shared through an external STOMP broker such as RabbitMQ or ActiveMQ, and the JWT secret to be provided through a secure vault rather than an environment variable.

Appendix A
Supplementary Material

This appendix contains supplementary material that is not required to understand the main text.

Bibliography

1. META PLATFORMS, INC. *React — A JavaScript library for building user interfaces* [online]. 2024. Available also from: `https://react.dev/`. [Accessed 3. 5. 2026].
2. VERCEL INC. *Next.js by Vercel — The React Framework* [online]. 2024. Available also from: `https://nextjs.org/`. [Accessed 3. 5. 2026].
3. TAILWIND LABS INC. *Tailwind CSS — Rapidly build modern websites without ever leaving your HTML* [online]. 2024. Available also from: `https://tailwindcss.com/`. [Accessed 3. 5. 2026].
4. SHADCN. *shadcn/ui — Beautifully designed components that you can copy and paste into your apps* [online]. 2024. Available also from: `https://ui.shadcn.com/`. [Accessed 3. 5. 2026].
5. ORACLE CORPORATION. *Java Platform, Standard Edition 21* [online]. 2023. Available also from: `https://docs.oracle.com/en/java/javase/21/`. [Accessed 3. 5. 2026].
6. BROADCOM INC. *Spring Boot* [online]. 2024. Available also from: `https://spring.io/projects/spring-boot`. [Accessed 3. 5. 2026].
7. THE POSTGRESQL GLOBAL DEVELOPMENT GROUP. *PostgreSQL: The World's Most Advanced Open Source Relational Database* [online]. 2024. Available also from: `https://www.postgresql.org/`. [Accessed 3. 5. 2026].
8. BROADCOM INC. *Spring Security* [online]. 2024. Available also from: `https://spring.io/projects/spring-security`. [Accessed 3. 5. 2026].
9. TEAM, SockJS. *SockJS-client* [online]. 2024. Available also from: `https://github.com/sockjs/sockjs-client`. [Accessed 3. 5. 2026].
10. META PLATFORMS, INC. *Jest — Delightful JavaScript Testing* [online]. 2024. Available also from: `https://jestjs.io/`. [Accessed 10. 6. 2026].

11. VITEST TEAM. *Vitest — A blazing fast unit test framework powered by Vite* [online]. 2024. Available also from: `https://vitest.dev/`. [Accessed 10. 6. 2026].

12. CYPRESS.IO. *Cypress — JavaScript Component Testing and E2E Testing Framework* [online]. 2024. Available also from: `https://www.cypress.io/`. [Accessed 10. 6. 2026].

13. MICROSOFT CORPORATION. *Playwright — Fast and reliable end-to-end testing for modern web apps* [online]. 2024. Available also from: `https://playwright.dev/`. [Accessed 10. 6. 2026].

14. GOOGLE LLC. *Google GenAI Java SDK* [online]. 2024. Available also from: `https://github.com/googleapis/java-genai`. [Accessed 28. 4. 2026].

15. APACHE SOFTWARE FOUNDATION. *Apache PDFBox — A Java PDF Library* [online]. 2024. Available also from: `https://pdfbox.apache.org/`. [Accessed 28. 4. 2026].

Contents of Enclosed Media

```
/
├── README.md ... brief description of the repository and setup instructions
├── backend ..... directory containing the backend Spring Boot application source code
├── frontend . directory containing the frontend Next.js application source code
├── documentation ......... directory containing the thesis documentation
│   ├── text ............................. LaTeX source files of the thesis
│   └── ctufit-thesis.pdf ........... the final thesis text in PDF format
└── docker-compose.yml .... configuration file for running the application locally
```